ART &CULTURE

17. The well-known painting "Bani Thani" belongs to the

(a) Bundi school (b) Jaipur school
(c) Kangra school (d) Kishangarh school

Ans.(d) Bani-Thani is an Indian miniature painting painted by Nihal Chand from the Marwar school of Kishangarh. The painting's subject, Bani Thani was a singer and poet in Kishangrh in time of the king Sawant Singh. (1748-1764). She has been compared with the Mona Lisa. This painting was featured in an Indian stamp issued in 1973.

18. With reference to the cultural history of India, consider the following statements :

1. **White marble was used in making Buland Darwaza and Khankah at Fatehpur Sikri.**
2. **Red sandstone and marble were used in making Bara Imambara and Rumi Darwaza at Lucknow.**

Which of the statements given above is/are correct ?

(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

Ans. (d): The Buland Darwaza at Fatehpur Sikri was built in 1602 AD by the great Mughal Emperor Akabar to commemorate victory over Gujarat. It is made of red and buff sandstone, decorated by white and black marble and higher than the mosque's courtyard. Therefore statement (1) is incorrect. Asaf-ud Daula rule saw a devastating famine, which created an economic crisis, a food-for-work programme was started. The famous Asafi Imambara or Bara Imambara of Lucknow, was built to give employment and revenue to the public. Instead of stones and marble, brick and lime were used. Therefore, statement (2) is also incorrect.

19. Consider the following pairs :

Tradition		State
1. Chapchar Kut festival	–	Mizoram
2. Khongjom Parba ballad	–	Mainpur
3. Thang-Ta dance	–	Sikkim

Which of the pairs given above is/are correct?

(a) 1 only (b) 1 and 2
(c) 3 only (d) 2 and 3

Ans. (b): The statement in question (1) and (2) are correctly matched while statement (3) is incorrect as Thang-Ta dance is an ancient martial art form of Manipur. Chapchar Kut is an agricultural festival celebrated in February- March in Mizroam while Khongjom Parba is a singing style of the ballad song of Manipur.

INDIAN GEOGRAPHY

• Physical Structure, Drainage System, Human Geography

20. Which one of the following is an artificial lake?

(a) Kodaikanal
(b) Kolleru (Andhra Pradesh)
(c) Nainital (Uttarakhand)
(d) Renuka (Himachal Pradesh)

Ans. (a): Kodaikanal Lake, also known as Kodi Lake is a manmade lake located in the Kodaikanal city in Dindigual district in Tamil Nadu, India. Sir Vere Henry Levinge the then collector of Madurai, was instrumental in creating the lake in 1863, amidst the Kodikonal town which was developed by the British and early missionaries from USA.

21. Consider the following statements :

1. **The Barren Island volcano is an active volcano located in the Indian territory.**
2. **Barren Island lies about 140 km east of Great Nicobar.**
3. **The last time the Barren Island volcano erupted was in 1991 and it has remained inactive since then.**

Which of the statements given above is/are correct ?

(a) 1 only (b) 2 and 3
(c) 3 only (d) 1 and 3

Ans. (c): Barren Island volcano is not active and it is located 140 kms to the east of Port Blair (lies in South Andaman), while great Nicobar is the southernmost region of Andaman & Nicobar and lies further South of Port Blair.
The Barren Island volcano had been lying dormant for more than 150 years until it saw a major eruption in 1991. Since, then it has shown intermittent activity, including eruptions in 1995, 2005 and 2017.

• Monsoon, Forest, Soil, Irrigation, Projects, Agriculture

22. As per the NSSO 70th Round "Situation Assessment Survey of Agricultural Households", consider the following statements:

1. **Rajasthan has the highest percentage share of agricultural households among its rural households.**
2. **Out of the total agricultural households in the country, a little over 60 percent belong to OBCs.**
3. **In Kerala, a little over 60 percent of agricultural households reported to have received maximum income from sources other than agricultural activities.**

Which of the statements given above is/are correct?

(a) 2 and 3 only (b) 2 only
(c) 1 and 3 only (d) 1, 2 and 3

Ans. (c): Rajasthan has the highest agricultural households (78.4%) among rural households, in terms of proportion, in absolute numbers, it is Uttar Pradesh.
The share of OBC households among rural households is about 45 percent.
In Kerala, about 61 percent of agricultural families have more income from agricultural sources.
Therefore, the correct answer is option (c).

23. Which of the following is/are the possible consequence/s of heavy sand mining in riverbeds ?

1. **Decreased salinity in the river**
2. **Pollution of groundwater**
3. **Lowering of the water-table**

Select the correct answer using the code given below :

(a) 1 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

Ans. (b): The removal of sand from the river bed increases the flowing water's velocity, with the distorted flow regime eventually eroding the river banks.
The sand acts like a sponge, which helps in recharging the water table; its progressive depletion in the river is accompanied by declining water tables in the nearby areas. Depletion of sand in the stream bed causes the deepening of rivers and estuaries and the enlargement of river mouths and coastal inlets. It leads to saline water incursion.
The sand acts as an efficient filter for various pollutants and thus maintains the quality of water in rivers and other aquatic ecosystems.
Hence, the correct answer is option (b).

24. With reference to agricultural soils, consider the following statements :

1. **A high content of organic matter in soil drastically reduces its water holding capacity.**
2. **Soil does not play any role in the sulphur cycle.**
3. **Irrigation over a period of time can contribute to the salinization of some agricultural lands.**

Which of the statements given above is/are correct ?

(a) 1 and 2 only (b) 3 only
(c) 1 and 3 only (d) 1, 2 and 3

Ans. (b): Statement 1 is wrong because it increases water holding capacity. A higher organic matter content in the soil will result in more humus and an increase in its water holding capacity.
Statement 2 is incorrect because water plays a role in the Sulphur cycle. Sulphur, like nitrogen and carbon is an essential part of living matter because sulphur - containing amino acids are always present in almost all kinds of proteins. Plants can absorb directly the sulphur containing amino acids but these amino acids fulfill only a small proportion or requirements of plants. Sulphur passes through the cycle of transformation mediated by microorganisms. It accumulates in the soil mainly as a constituent of organic compounds and has to be converted to sulphates to become readily available to plants.
Irrigation causes organic matter to leach and making land saline over-irrigation can lead to salinity in soils because of over-use of ground-water and/or the rise of the water table.
So, correct answer is option (b).

• Minerals, Energy Resources, Industry and Trade

25. Consider the following statements:

1. **In India, State Governments do not have the power to auction non-coal mines.**
2. **Andhra Pradesh and Jharkhand do not have gold mines.**
3. **Rajasthan has iron ore mines.**

Which of the statements given above is/are correct?

(a) 1 and 2 (b) 2 only
(c) 1 and 3 (d) 3 only

Ans. (d): As per the new mining law Mines and Minerals (Development and Regulation) Amendment Act, 2015- which came into effect in January 2015, the non-coal mines have to be auctioned by the respective state governments.

- Jharkhand has both alluvial and native gold. Alluvial gold is obtained from the sands of the Subarnarekha river, Sona Nadi in Singhbhum district and the streams draining the Sonapat valley. Native gold is found near Lawa in the Singhbhum district.
- In Andhra Pradesh, gold deposits are found in Ranagiri in Anantapur district.
- In Rajasthan, the iron-ore deposits are located in the districts of Jaipur, Udaipur, Jhunjhuna, Sikar, Bhilwara, Alwar, Bharatpur, Dausa and Banswara.

So the correct answer is option (d).

WORLD GEOGRAPHY

• Universe and Solar System

26. Consider the following statements:

1. **The Earth's magnetic field has reversed every few hundred thousand years.**
2. **When the Earth was created more than 4000 million years ago, there was 54% oxygen and no carbon dioxide.**
3. **When living organisms originated, they modified the early atmosphere of the Earth.**

Which of the statements given above is/are correct ?

(a) 1 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

Ans. (c): Statement 1- Reversal in Earth's magnetic field is called geomagnetic reversal. It occurs every few million years. It is a change in a planet's magnetic field such that the positions of magnetic north and magnetic south are interchanged, while geographic north and geographic south remain the same. Thus, the statement 1 is correct.
Statement 2- Early atmosphere has an abundance of hydrogen sulphide and carbon dioxide. There were smaller proportions of water vapour, ammonia and methane. The present composition got finalized very late around, 600 million years ago (Cambrian period). Volcanism and the very evolution of life played an important role in changing the atmospheric composition. Thus statement (2) is incorrect.
Statement 3- Respiration and microbial action by organisms modified the early atmosphere. Thus, this statement is correct.
So correct answer is option (c)

• Hydrosphere

27. Which of the following has/have shrunk immensely/dried up in the recent past due to human activites?